FORMATIVE ASSESSMENT-2
PROJECT WORK

1. Name of the Project

Properties of Acids, Bases and Salts

2. Purpose of the Project

The primary purpose of this project is to systematically investigate the physical and chemical properties of acids and bases. This involves understanding their reactions with metals, observing color changes with various natural and synthetic indicators, analyzing the concept of the pH scale, and studying the formation and real-life utility of different chemical salts.

3. Hypothesis

It is hypothesized that acids will exhibit a sour taste and turn blue litmus paper red, while bases will have a soapy touch and turn red litmus blue. Furthermore, reacting an acid with an active metal will produce hydrogen gas (observable via a "pop" sound), and the reaction between an acid and a base will result in a neutralization reaction forming a neutral salt and water.

4. Materials Used

- **Chemicals:** Dilute Hydrochloric Acid (HCl), Dilute Sodium Hydroxide ($NaOH$), Zinc granules, Sodium Carbonate (Na_2CO_3), Phenolphthalein, Methyl Orange.
- **Apparatus:** Test tubes, test tube stand, delivery tube, one-holed rubber cork, beaker, matchbox, and candle.

- **Indicators:** Red and Blue Litmus paper, pH paper with a universal indicator color chart.

5. Procedure

1. **Litmus Test:** Take small quantities of the given samples (Acid and Base) in separate watch glasses. Dip a strip of red litmus paper and blue litmus paper into each and record the color change.
2. **Reaction with Metals:** Take about 5ml of dilute HCl in a test tube and add a few pieces of zinc granules to it. Pass the gas being evolved through soap solution using a delivery tube. Bring a burning candle near the soap bubbles filled with the gas and observe.
3. **Neutralization Reaction:** Take 2ml of dilute $NaOH$ solution in a test tube and add two drops of phenolphthalein indicator. Add dilute HCl solution to it drop by drop until a color change is observed.

6. Data Collection

The following observations were recorded when testing different substances with various chemical indicators:

Sample Solution	Red Litmus Paper	Blue Litmus Paper	Phenolphthalein	Methyl Orange
Dilute HCl	No change (Red)	Turns Red	Colorless	Red
Dilute $NaOH$	Turns Blue	No change (Blue)	Pink	Yellow
Lemon Juice	No change	Turns Red	Colorless	Red
Baking Soda Solution	Turns Blue	No change	Pink	Yellow

7. Introduction

Chemical compounds are broadly classified into three categories: acids, bases, and salts. This classification is based on their distinct chemical properties and how they interact with each other and other elements. According to the Arrhenius concept, acids are substances that release hydrogen ions (H^+) in an aqueous solution, whereas bases release hydroxyl ions (OH^-). To measure the strength of these acids and bases, the **pH scale** is utilized, which ranges from 0 to 14.

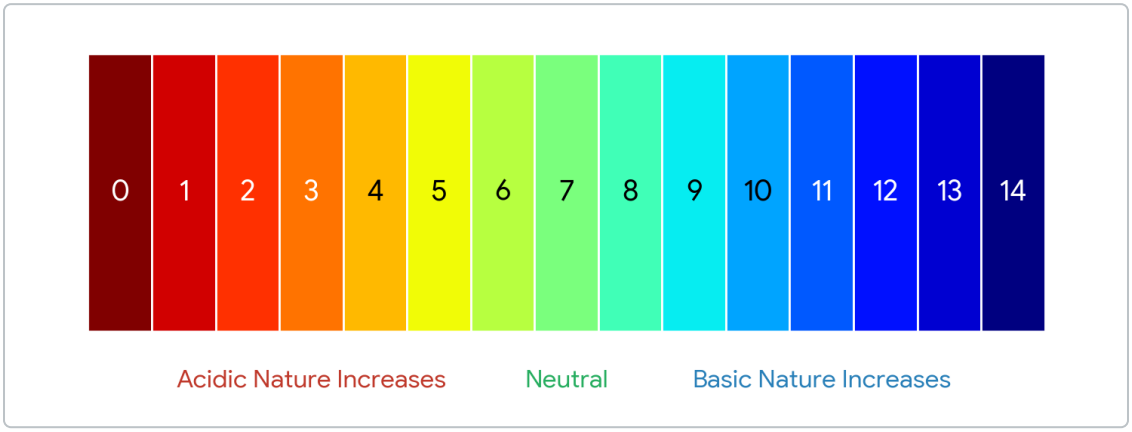


Image 1: The pH Scale indicating Acidic, Neutral, and Basic Ranges

8. Acids

The term 'acid' is derived from the Latin word 'acidus', which means sour. Acids are fundamentally corrosive in nature and are found in many natural substances like citrus fruits (citric acid) and curd (lactic acid).

Sub-topic A: Physical Properties

- They have a distinct sour taste.
- They turn blue litmus paper red.
- Aqueous solutions of acids conduct electricity due to the presence of free ions.

Sub-topic B: Chemical Properties

- **Reaction with Metals:** Acids react with active metals like Zinc and Magnesium to form metal salts and liberate Hydrogen gas.

  $$Zn + 2HCl \rightarrow ZnCl_2 + H_2 \uparrow$$

- **Reaction with Carbonates:** They react with metal carbonates to liberate Carbon Dioxide gas with brisk effervescence.

9. Bases

Bases are chemically opposite to acids. Bases that are soluble in water are specifically known as alkalis (e.g., Sodium Hydroxide, Potassium Hydroxide).

Sub-topic A: Physical Properties

- They have a bitter taste and feel soapy or slippery to the touch.
- They turn red litmus paper blue.
- Like acids, aqueous solutions of bases also conduct electricity.

Sub-topic B: Chemical Properties

- **Reaction with Non-metal Oxides:** Bases react with non-metal oxides (which are acidic in nature) to form salt and water, similar to a neutralization reaction.
- **Reaction with Metals:** Strong bases react with certain metals like Zinc to produce Hydrogen gas, forming complex salts like Sodium Zincate (Na_2ZnO_2).

10. Salts

A salt is an ionic compound formed by the neutralization reaction between an acid and a base. Salts consist of a positive ion (cation) from the base and a negative ion (anion) from the acid. While neutral salts (derived from strong acids and strong bases) have a pH of 7, acidic or basic salts can form depending on the strengths of the reacting components.

Common salt (Sodium Chloride, $NaCl$) serves as an essential raw material for producing several chemical compounds crucial for everyday life:

- **Bleaching Powder ($CaOCl_2$):** Used for disinfecting drinking water and bleaching cotton.
- **Baking Soda ($NaHCO_3$):** Used in cooking to make cakes fluffy and as an antacid.
- **Washing Soda ($Na_2CO_3 \cdot 10H_2O$):** Used in glass, soap, and paper industries.
- **Plaster of Paris ($CaSO_4 \cdot \frac{1}{2}H_2O$):** Used by doctors for supporting fractured bones.

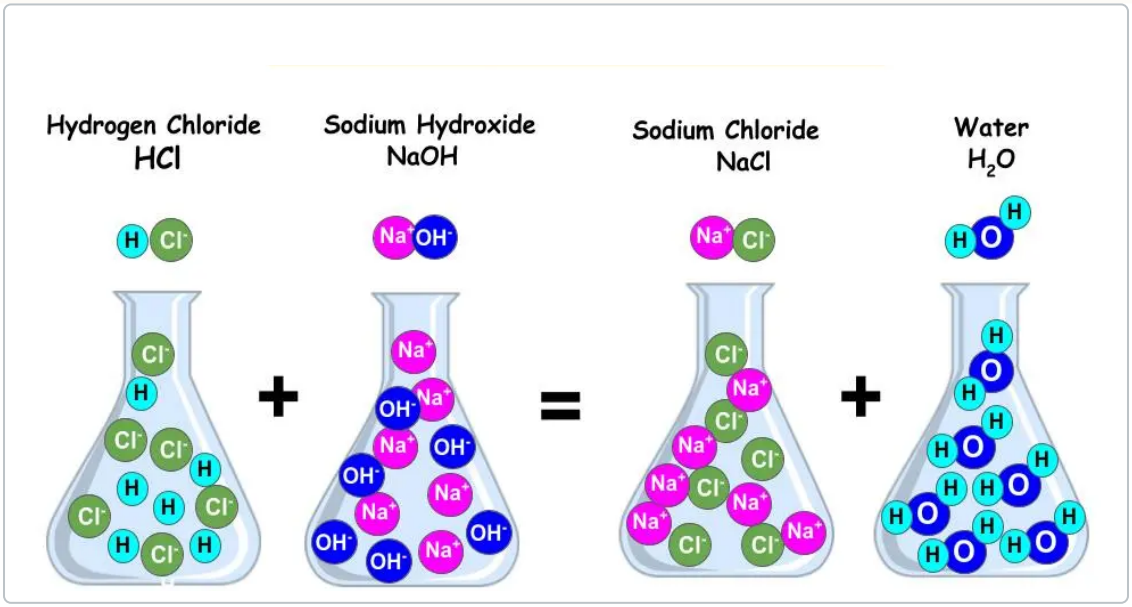


Image 2: Neutralization Process and Important Salts Derived from NaCl

11. Observation

During the experimentation phase, several distinct phenomena were observed. When Zinc granules were added to the dilute Hydrochloric Acid, colorless bubbles of gas were vigorously formed. Upon passing this gas through a soap solution and bringing a burning candle near the bubbles, the gas burned with a distinct "pop" sound, confirming it as Hydrogen gas.

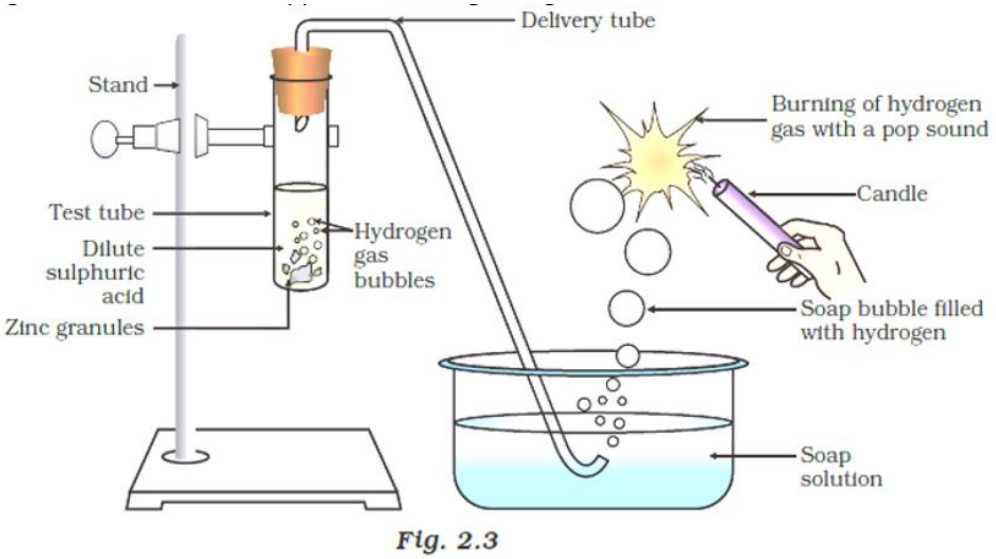


Fig. 2.3

Image 3: Testing the Evolution of Hydrogen Gas (Pop Sound)

In the neutralization experiment, the phenolphthalein indicator turned pink in the presence of the base ($NaOH$). As dilute HCl was added drop by drop, the pink color eventually disappeared, indicating that the base was completely neutralized by the acid, bringing the solution to a neutral pH.

12. Conclusion

The project successfully validates the formulated hypothesis. Acids and bases possess distinctly identifiable physical properties (taste, touch) and chemical behaviors (litmus color changes). The evolution of Hydrogen gas upon reacting an acid with a metal was experimentally proven. Furthermore, the combination of acids and bases results in a neutralization reaction that yields salts. These salts are incredibly diverse and form the foundation for numerous industrial applications and daily household uses.